# COMP9334: Capacity Planning of Computer Systems and Networks

Optimisation (3):
Network flow

# The story so far

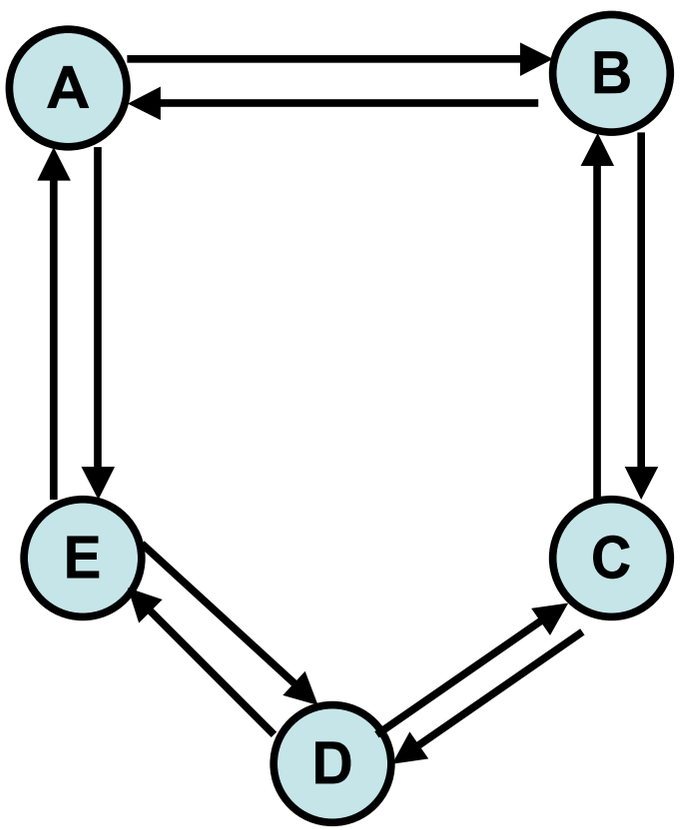
- Linear programming (LP)
  - Real values for decision variables, linear in objective function, linear in constraints
  - Large LP problems can be solved routinely
- Integer programming (IP)
  - Some decision variables can only take integer values
  - Some decision variables can only take binary values, e.g. for making yes-or-no decisions
  - IP problems can be solved using branch and bound in principle
  - Computation complexity is generally exponential except for unimodular problems
  - Use linear integer programming whenever possible

# Week 9B

- Applications of integer programming for network flow problems
  - Traffic Engineering
  - Dimensioning problem
  - Topology design

# Traffic Engineering Example (1)

An ISP owns the following network which connects
5 cities A, B, C, D and E.
Capacity of each link is 1000 Mbps

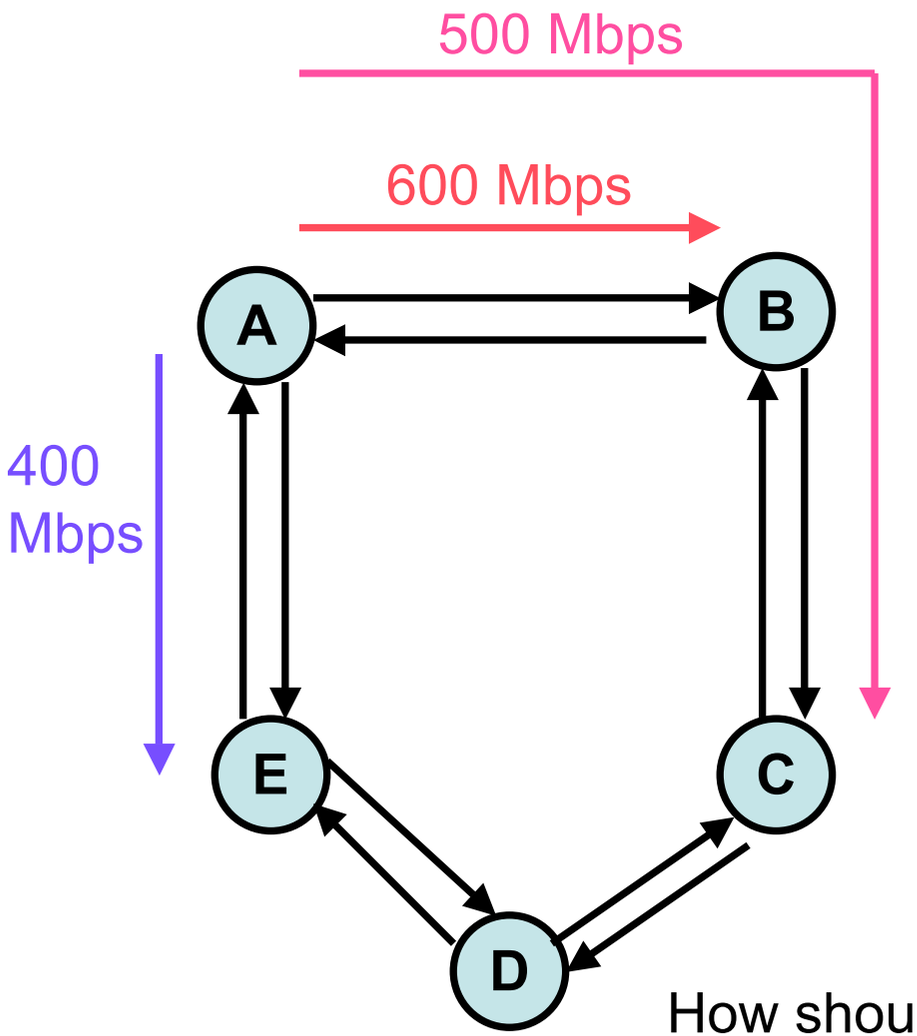



The traffic demands between cities are:
A to B: 600 Mbps
A to E: 400 Mbps
A to C: 500 Mbps

Question: How should we route the traffic so that the links are at most 80% utilised and we use the minimum amount of resources?

# Traffic Engineering Example (2)

Capacity of each link is 1000 Mbps




The traffic demands between cities are:
A to B: 600 Mbps
A to E: 400 Mbps
A to C: 500 Mbps

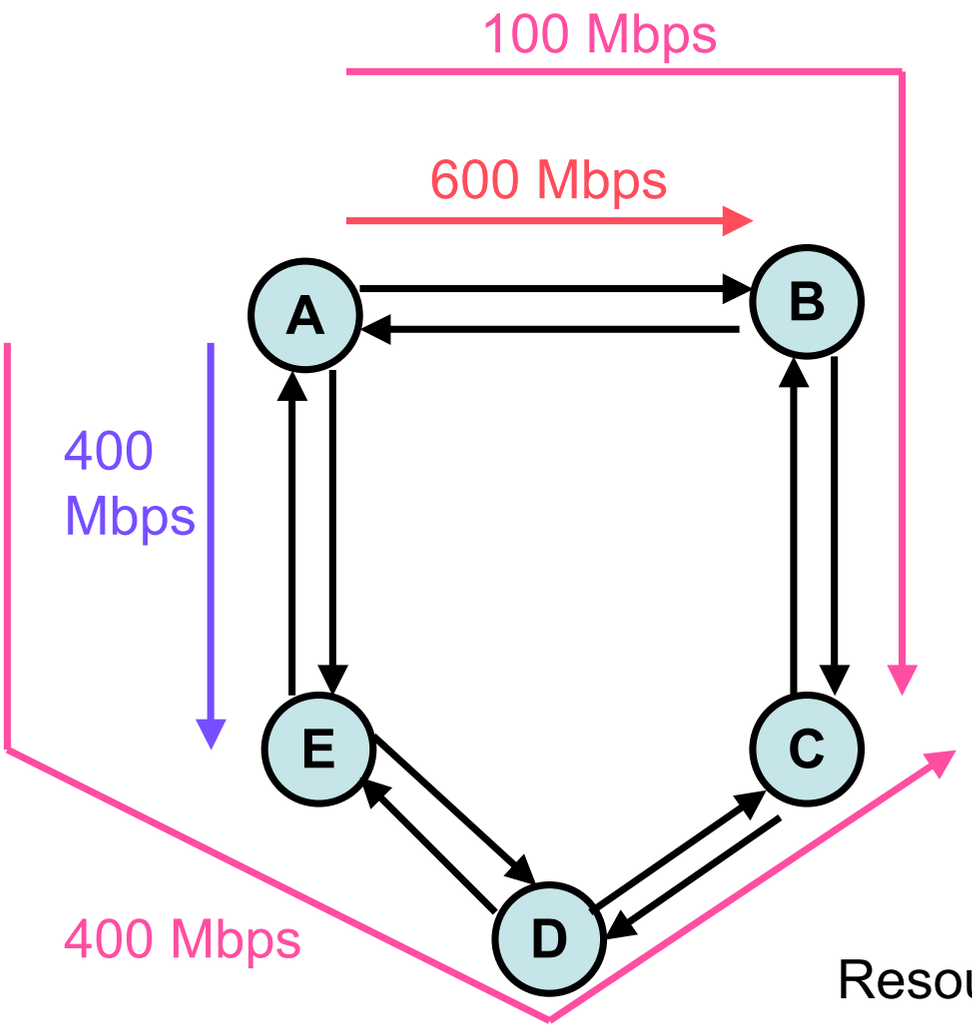
Let us assume that the traffic demand will take the shortest path between its end points

But 1100 Mbps in link A-B! The link is over-utilised!

How should you route the traffic?

# Traffic Engineering Example (3)

Capacity of each link is 1000 Mbps




- The traffic demands between cities are:
  - A to B: 600 Mbps
  - A to E: 400 Mbps
  - A to C: 500 Mbps

- Traffic in links
  - A-B = 700 Mbps
  - B-C = 100 Mbps
  - A-E = 800 Mbps
  - E-D = 400 Mbps
  - D-C = 400 Mbps

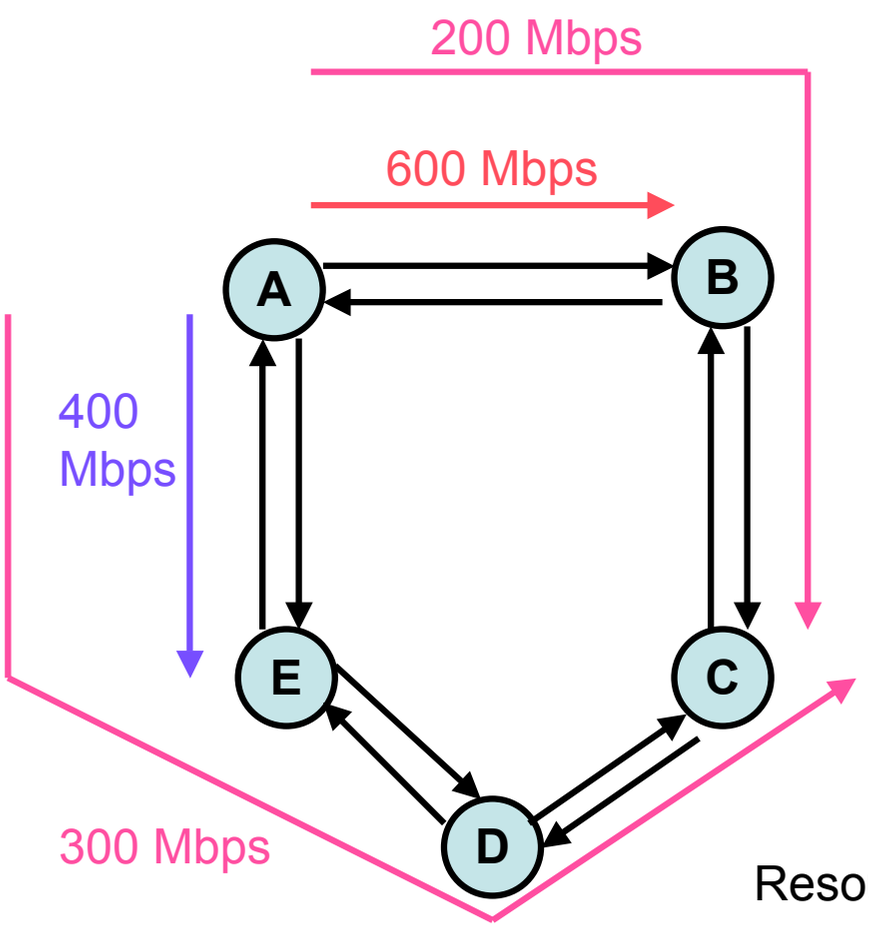
} +

- Link A-E is 80% utilised. Others are less utilised.

Resources used = 2400Mbps

# Traffic Engineering Example (4)

Capacity of each link is 1000 Mbps




- The traffic demands between cities are:
  - A to B: 600 Mbps
  - A to E: 400 Mbps
  - A to C: 500 Mbps

- Traffic in links
  - A-B = 800 Mbps
  - B-C = 200 Mbps
  - A-E = 700 Mbps
  - E-D = 300 Mbps
  - D-C = 300 Mbps

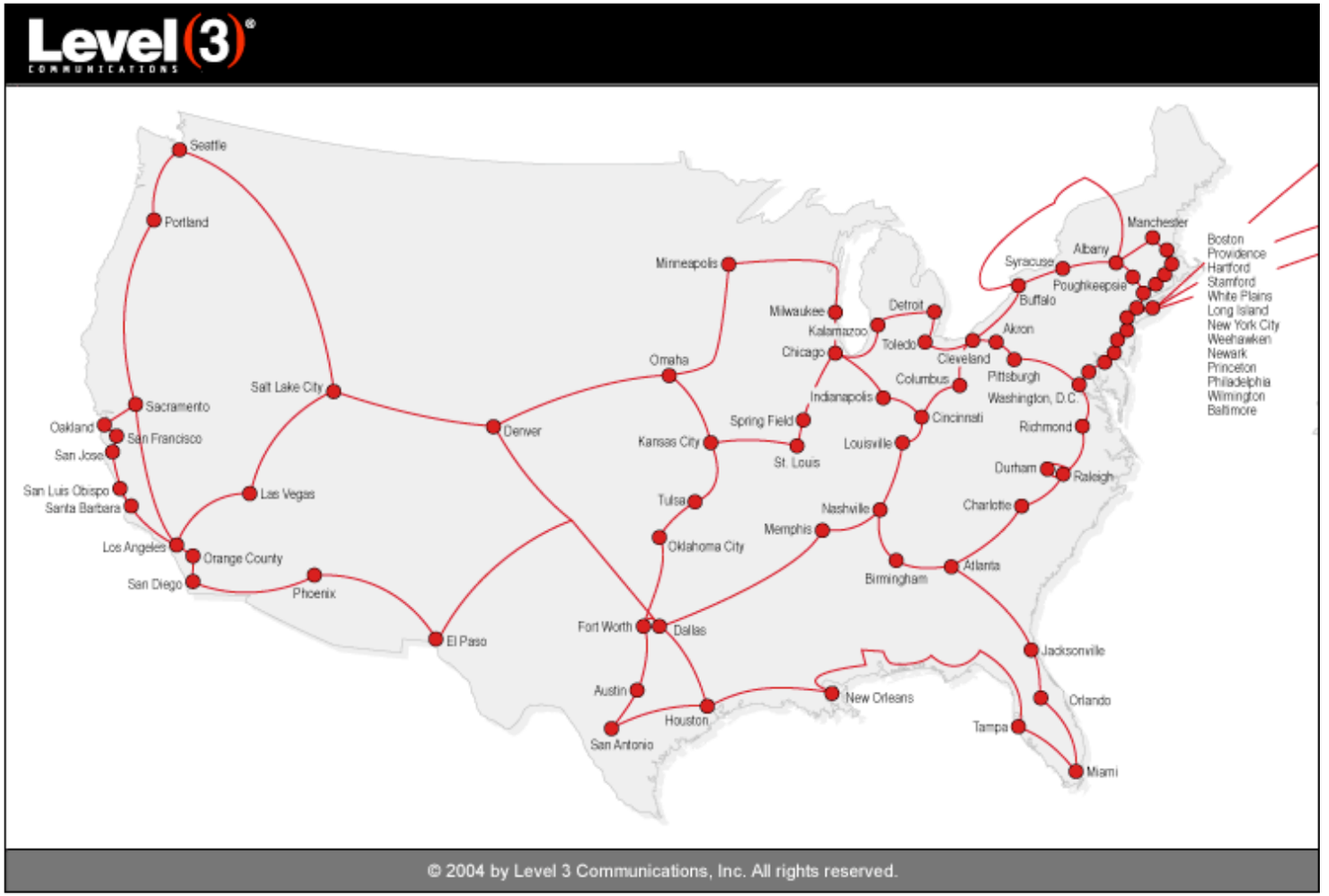
} +

- Link A-B is 80% utilised. Others are less utilised.

Resources used = 2300Mbps

# Traffic Engineering

- General traffic engineering problem:
  - Given:
    - A network (i.e. nodes, links and their capacites)
    - The traffic demand between each pair of nodes.
  - Find: how to route the traffic to best utilise the resource
- The traffic engineering example earlier was simple, but for a commercial carrier (Next slide shows the network map of a commercial carrier.), it's no longer so.
  - Traffic engineering problems can be solved systematically using integer programming
  - These problems are generally known as network flow problems. *Note: flow is synonymous with traffic demand between a pair of nodes.*
  - We will start with the simplest network flow problem, finding the shortest path for one flow.


Level(3)
COMMUNICATIONS
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Milwaukee
Chicago
Kalamazoo
Detroit
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Cleveland
Akron
Buffalo
Syracuse
Albany
Manchester
Poughkeepsie
Boston
Providence
Hartford
Stamford
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Long Island
New York City
Weehawken
Newark
Princeton
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Wilmington
Baltimore
Columbus
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Washington, D.C.
Indianapolis
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St. Louis
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Birmingham
Atlanta
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Tampa
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New Orleans
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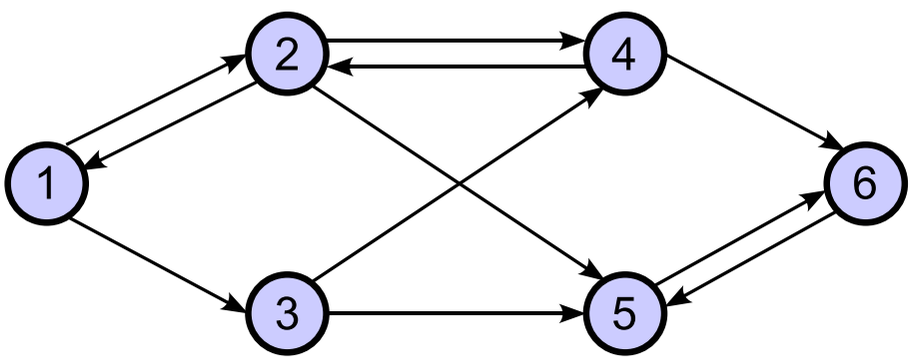
# Network flow problems

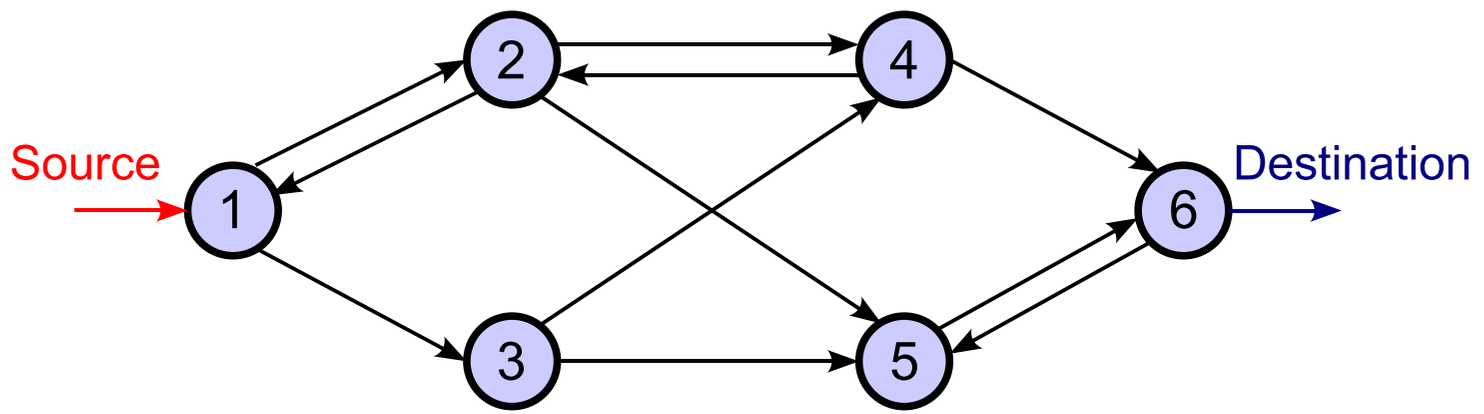
- Network flow problems are important applications of integer programming
  - Move some entity from one point to another in the network
  - Given alternative ways, find the most efficient one, e.g. minimum cost, maximum profit, etc.

- Network is represented as a directed graph $G = (N, E)$
  - $N$ = the set of nodes, e.g. $N = \{1, 2, 3, 4, 5, 6\}$
  - $E$ = the set of directed edges, e.g. $E = \{\langle 1, 2\rangle, \langle 1, 3\rangle, \langle 2, 1\rangle, \ldots\}$



# Finding the shortest path

- Aim: Find the shortest path from the source node to the destination node of a flow




- Cost is generally assumed to be additive, i.e. cost of a path = sum of the cost of using each edge in the path
  - E.g. cost of using edges $\langle 1, 2 \rangle$, $\langle 2, 4 \rangle$ and $\langle 4, 6 \rangle$
    = cost of edge $\langle 1, 2 \rangle$ + cost of edge $\langle 2, 4 \rangle$ + cost of edge $\langle 4, 6 \rangle$

# Shortest path problem (SPP)

- Given
  - A directed graph $G = (N, E)$
  - A flow of size $1$ enters at node $s$ (source) and leaves at node $d$ (destination)
  - It costs $c_{i,j}$ for using directed edge $\langle i, j \rangle$
- Find which directed edges the flow should use in order that
  - The total cost is minimized
  - The entire flow must use only one path
- Logical decision: Should I use a directed edge or not?

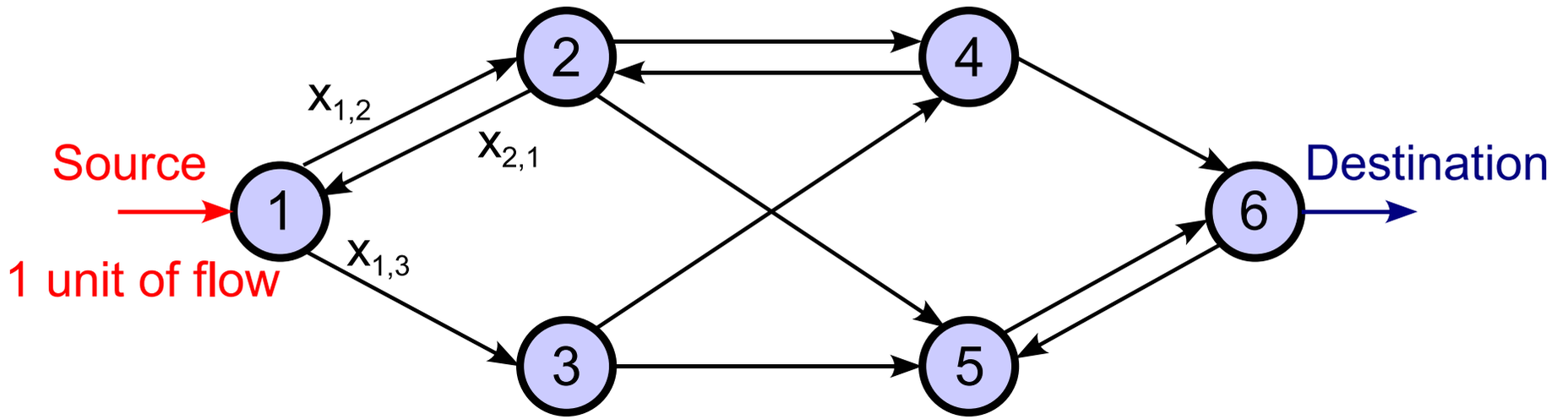
# Formulating SPP

- Decision variables

$$x_{i,j} = \begin{cases} 1 & \text{if directed edge } \langle i, j \rangle \text{ in } E \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

- We assume 1 unit of flow from the source to the destination
- An important part of the formulation is to make sure the directed edges selected actually form a connected path from the source to the destination
  - This is by adding the conservation of flow constraints

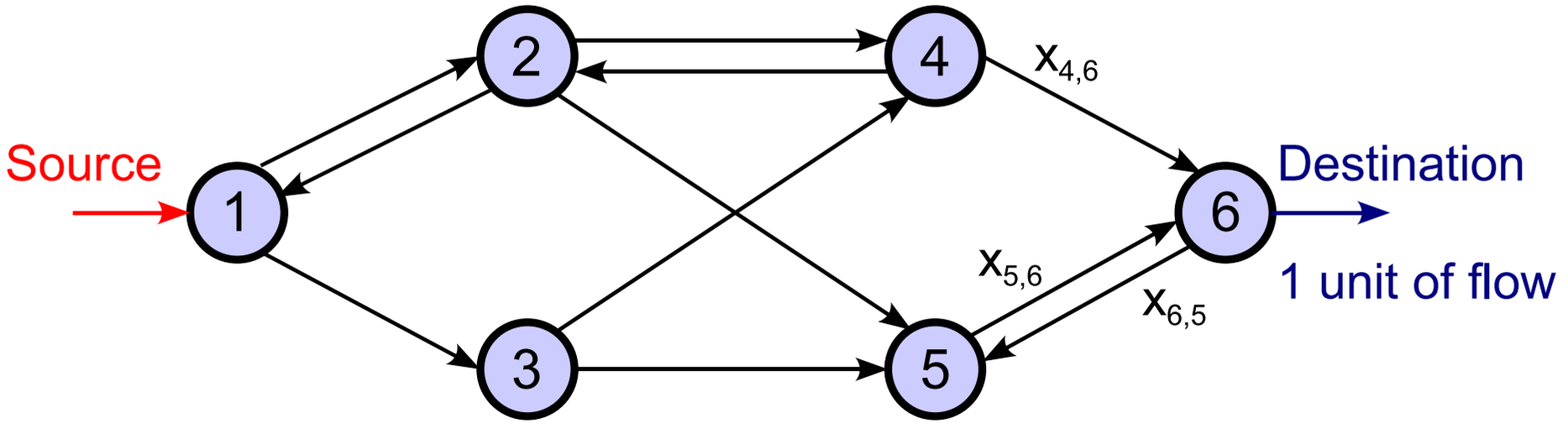
# Conservation of flow: Source node




- What goes in = What goes out
  - Flow going into node 1 from external = 1
  - Flow going into node 1 from neighboring nodes = $x_{2,1}$
    - Second index is “1”
  - Flow going from node 1 to neighboring nodes = $x_{1,2} + x_{1,3}$
    - First index is “1”
  - Therefore, we have

$$1 + x_{2,1} = x_{1,2} + x_{1,3}$$

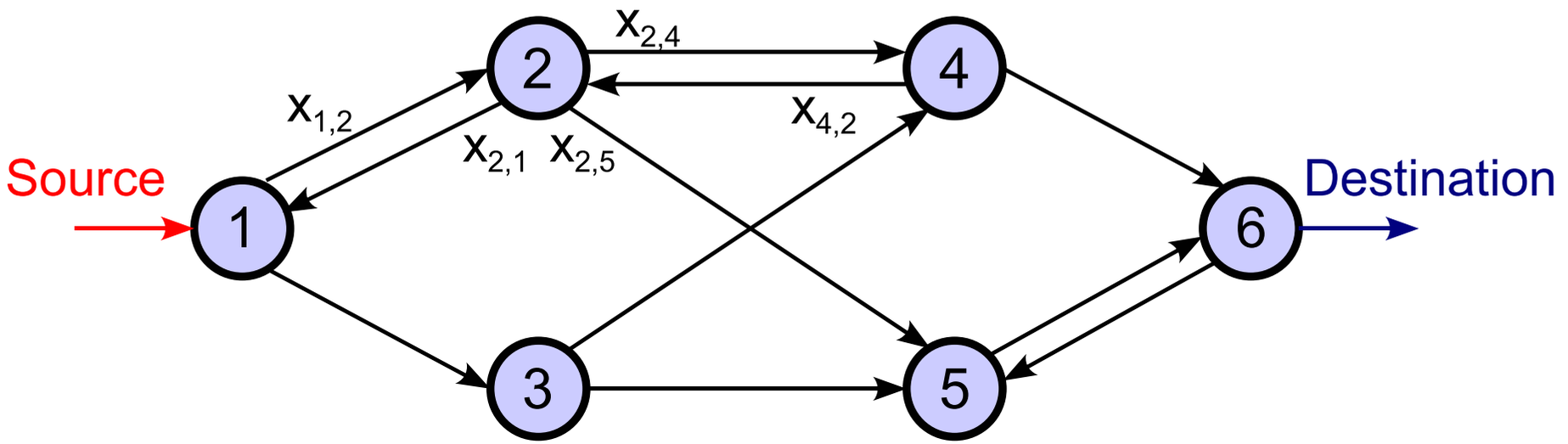
# Conservation of flow: Destination node




- What goes in = What goes out
  - Flow going into node 6 from neighboring nodes = $x_{4,6} + x_{5,6}$
    - Second index is “6”
  - Flow going from node 6 to neighboring nodes = $x_{6,5}$
    - First index is “6”
  - Flow going from node 6 to external = 1
  - Therefore, we have

$$x_{4,6} + x_{5,6} = x_{6,5} + 1$$

# Conservation of flow: Other nodes




- E.g. flow conservation at node 2: What goes in = What goes out
  - Flow going into node 2 from neighboring nodes = $x_{1,2} + x_{4,2}$
    - Second index is “2”
  - Flow going from node 2 to neighboring nodes = $x_{2,1} + x_{2,4} + x_{2,5}$
    - First index is “2”
  - Therefore, we have

$$x_{1,2} + x_{4,2} = x_{2,1} + x_{2,4} + x_{2,5}$$

# Conservation of flow constraints

- In our example network, the source is node 1, so the constraint is

$$1 + x_{2,1} = x_{1,2} + x_{1,3}$$

  - This can be rewritten as

$$\sum_{j:\langle 1,j\rangle \in E} x_{1,j} - \sum_{j:\langle j,1\rangle \in E} x_{j,1} = 1$$

- The destination is node 6, so the constraint is

$$x_{4,6} + x_{5,6} = x_{6,5} + 1$$

  - This can be rewritten as

$$\sum_{j:\langle 6,j\rangle \in E} x_{6,j} - \sum_{j:\langle j,6\rangle \in E} x_{j,6} = -1$$

# Conservation of flow constraints (cont.)

- For all other nodes (neither a source or a destination), e.g. node 2, the constraint is

$$x_{1,2} + x_{4,2} = x_{2,1} + x_{2,4} + x_{2,5}$$

  - This can be rewritten as

$$\sum_{j:\langle 2,j\rangle \in E} x_{2,j} - \sum_{j:\langle j,2\rangle \in E} x_{j,2} = 0$$

- The flow conservation constraints can be written in a compact form

$$\sum_{j:\langle i,j\rangle \in E} x_{i,j} - \sum_{j:\langle j,i\rangle \in E} x_{j,i} = 0, \quad i \in N - \{s, d\}$$

# IP formulation for SPP

- SPP can be formulated as

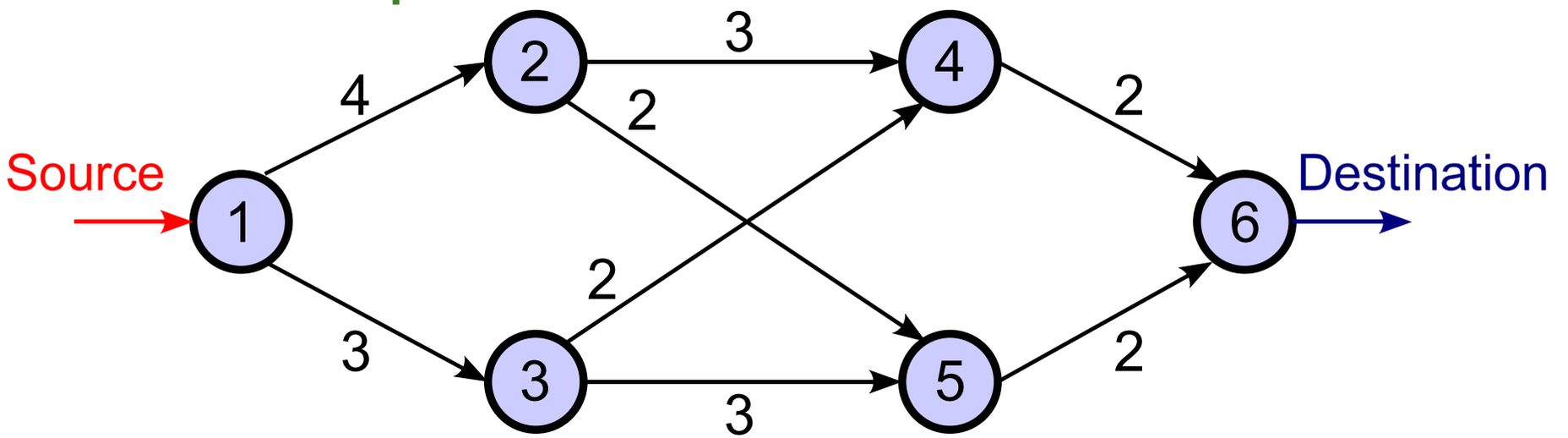
$$\min \sum_{\langle i,j \rangle \in E} c_{i,j} x_{i,j}$$

subject to

$$\sum_{j:\langle i,j \rangle \in E} x_{i,j} - \sum_{j:\langle j,i \rangle \in E} x_{j,i} = \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \in N - \{s, d\} \\ -1 & \text{if } i = d \end{cases}$$

$$x_{i,j} \in \{0, 1\} \text{ for all } \langle i, j \rangle \in E$$

# SPP example




- We will use AMPL/CPLEX for solving SPP in this example network
- Note:
  - It is far more efficient to use Dijkstra's algorithm for solving SPP
  - The reason of using integer programming here is for illustration only
- The files are `shortest.dat`, `shortest.mod` and `shortest_batch`

# Flow conservation constraints

- The flow conservation constraints ensure that there is only one path from the source node $s$ to destination node $d$

$$\sum_{j:\langle i,j\rangle \in E} x_{i,j} - \sum_{j:\langle j,i\rangle \in E} x_{j,i} = \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \in N - \{s, d\} \\ -1 & \text{if } i = d \end{cases}$$

$$x_{i,j} \in \{0, 1\} \text{ for all } \langle i, j\rangle \in E$$

# Introducing non-unit flow and link capacity (1)

(Note: A dot point preceded by ★ indicates that it is different from the setting of the shortest path problem.)

- Given
  - A directed graph $(N, E)$
  - ★ A flow of size $f$ with source node $s$ and destination node $d$
  - It costs $c_{ij}$ (per unit flow) for the flow to use directed edge $(i, j)$
  - ★ The capacity of the directed edge $(i, j)$ is $b_{ij}$
- Find which directed edges the flow should use in order that
  - The total cost is minimised
  - The entire flow must use only one path
  - ★ The flow on any directed edge does not exceed its capacity

# Introducing non-unit flow and link capacity (2)

- Decision variables are the same as before

$$x_{ij} = \begin{cases} 1 & \text{if directed edge } (i,j) \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

- The amount of flow on directed edge $(i,j)$ will be $fx_{ij}$

# Introducing non-unit flow and link capacity (3)

The problem formulation is

$$\min \sum_{(i,j)\in E} c_{ij}x_{ij}$$

subject to

$$\sum_{j:(i,j)\in E} x_{ij} - \sum_{j:(j,i)\in E} x_{ji} = \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \in N - \{s, d\} \\ -1 & \text{if } i = d \end{cases}$$

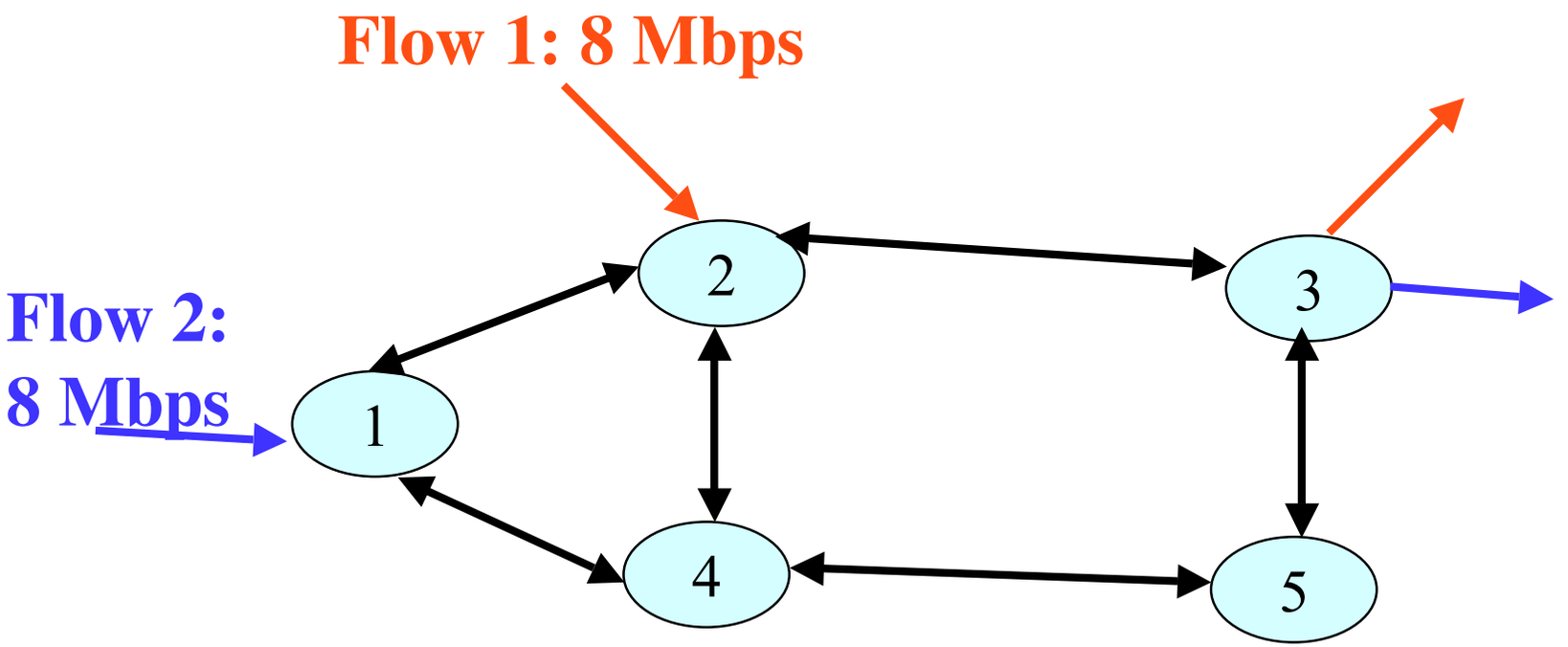
$$f x_{ij} \leq b_{ij} \text{ for all } (i,j) \in E \qquad (***)$$

$$x_{ij} \in \{0, 1\} \text{ for all } (i,j) \in E$$

Note: $(***)$ — this constraint ensures that only links with sufficient capacity may be chosen to carry the flow.
Solution: Eliminate edges with insufficient capacity, then Dijkstra.

# Multiple flows (1)

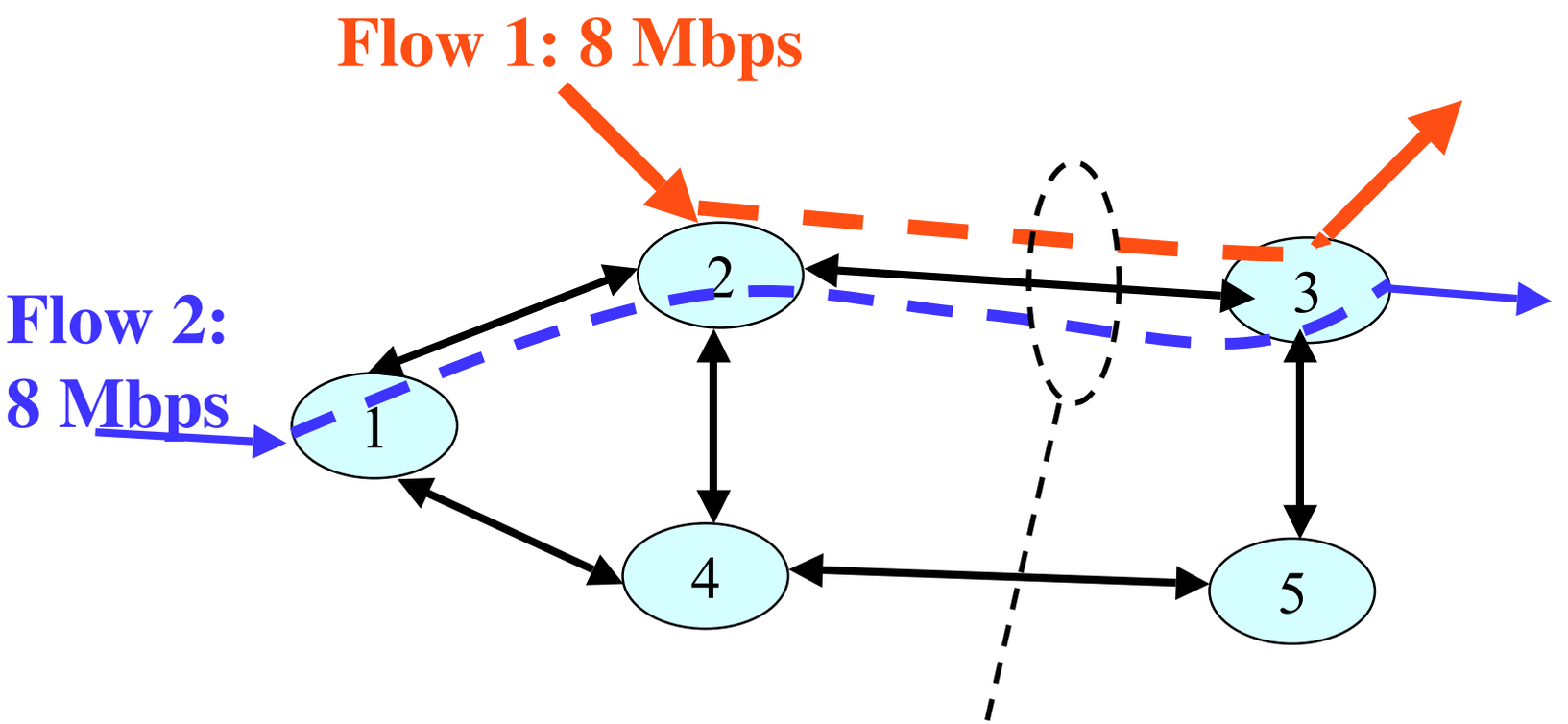
**Given: Each link has capacity of 10 Mbps**




**Assuming cost for each link is 1.**
**What if both flows use the shortest path?**

# Multiple flows (2)

**Given: Each link has capacity of 10 Mbps**




**16 Mbps of flows on 10Mbps**
**⇒ Utilisation > 1, High packet delay and loss**

# Traffic engineering problem

- Given
  - A directed graph $(N, E)$
  - ★ $m$ flows (indexed by $k = 1, 2, ...m$)
  - ★ Flow $k$ has size $f_k$, source node $s_k$, destination $d_k$
  - ★ It costs $c_{ij}$ for a unit of flow to use directed edge $(i, j)$
  - The capacity of directed edge is $b_{ij}$
- Find the directed edges that **each flow** should use in order that
  - The total cost is minimised
  - The entire flow must use only one path
  - ★ The *total flow* on a directed edge does not exceed its capacity

# Digression: Integral versus continuous traffic engineering

- There are two versions of traffic engineering problem
- The *integral* version where each flow must use only one path, i.e. all packets in a flow must use the same path
  - In order to ensure that the packets use a certain path, you can use source routing (available in IP version 6) or MPLS (multi-protocol label switching)
- The *continuous* version where each flow may use multiple paths, e.g. the one described on pages 5 – 6 of this lecture.
  - In order to split the flow, a classifier will be required at the router to send packets on different paths
- We will see how we can formulate the integral traffic engineering problem

# Traffic engineering IP formulation (1)

- Decision variables: $m$ sets of decision variables, one for *each flow*

$$x_{ijk} = \begin{cases} 1 & \text{if flow } k \text{ uses directed edge } (i,j) \\ 0 & \text{otherwise} \end{cases}$$

- The flow on directed edge $(i,j)$ will be

$$\sum_{k=1}^{m} f_k x_{ijk}$$

- Ex: $m = 3$. Flows 1 and 3 use edge (1,2) but flow 2 doesn't.

  Total flow in edge (1,2) = $f_1 + f_3$

  $\sum_{k=1}^{m} f_k x_{12k} = f_1 \times \underbrace{x_{121}}_{=1} + f_2 \times \underbrace{x_{122}}_{=0} + f_3 \times \underbrace{x_{123}}_{=1} = f_1 + f_3$

# Traffic engineering IP formulation (2)

The problem formulation is

$$\min \sum_{(i,j)\in E} \sum_{k=1}^{m} c_{ij} f_k x_{ijk}$$

subject to

$$\sum_{j:(i,j)\in E} x_{ijk} - \sum_{j:(j,i)\in E} x_{jik} = \begin{cases} 1 & \text{if } i = s_k \\ 0 & \text{if } i \in N - \{s_k, d_k\} \\ -1 & \text{if } i = d_k \end{cases} \quad k = 1, ..., m \ (*)$$

$$\sum_{k=1}^{m} f_k x_{ijk} \leq b_{ij} \quad \text{for all } (i,j) \in E \ (**)$$

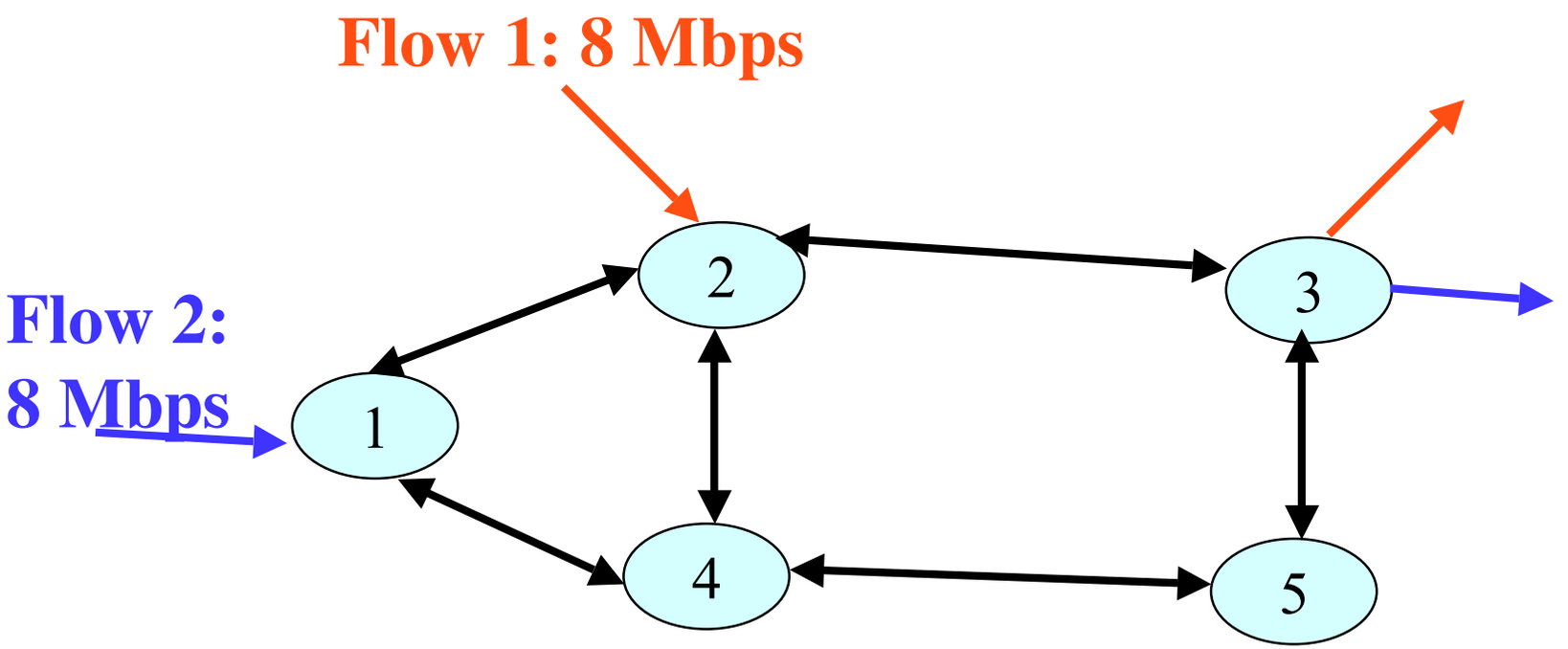
$$x_{ijk} \in \{0,1\} \text{ for all } (i,j) \in E,\ k = 1, ..., m$$

$(*)$ – One set of flow balance constraint per flow. Enforces flow $k$ is from $s_k$ to $d_k$

$(**)$ – Total flow on a link does not exceed its capacity.

# AMPL Example

**Given: Each link has capacity of 10 Mbps**




**Assuming cost for each link is 1.**
**What if both flows use the shortest path?**

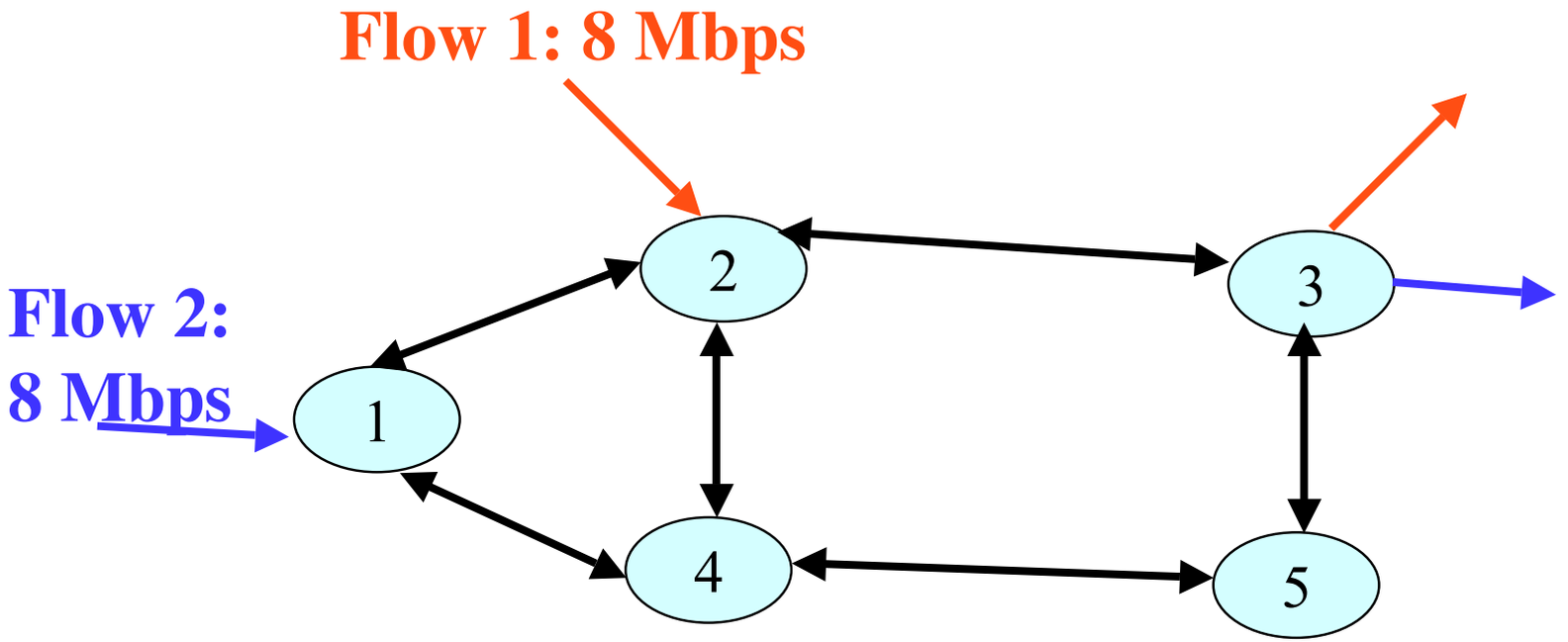
Files are `mcf1.dat`, `mcf1.mod` and `mcf1_batch`,

# Traffic engineering problem

- Also known as
  - The multi-commodity flow problem in operations research
  - Flow assignment problem
- Essence: assign a flow to a path so that performance is met
  - i.e. routing problem
- Many variations possible
  - Constraint on the path delay / number of hops
  - Constraint on packet loss rate

# Network design problem(1)

**In flow assignment, we assume the network topology and link capacities are given.**




**Why should we choose capacity 10Mbps? Why not 100Mbps? Why should we choose to have a link between (2,3) but not (2,5)?**

# Network design problem(2)

- Given
  - A set of nodes $N$
  - $m$ flows of size $f_k$, source $s_k$, destination $d_k$
  - Minimum network building cost
- Design options in network design problems
  - Topology: Which directed links to include
  - Capacity of the link
  - How the flows are routed?
- There are a few different network design problems

# Different network design problems

- Flow Assignment Problem
  - Given: flows, topology, capacity
  - Find: paths for the flows
- Capacity and Flow Assignment Problem
  - Given: flows, topology, network cost
  - Find: paths for the flows, capacity
- Topology, Capacity and Flow Assignment Problem
  - Given: flows, network cost
  - Find: paths for the flows, capacity, topology

# Summary

- Network flow problem
  - How to route traffic from a source node to destination node?
  - Many variations: From flow assignment only to topology, capacity and flow assignment
- Key new idea: Flow conservation constraint to enforce a path from a source node to a destination node

# References

- Advanced formulation of integer programming problems
  - Winston, “Operations Research”, Section 9.2
- Network flow problems
  - Ahuja et al, “Network Flows”, Sections 1.2